since if they are open there is no topologically stable notion of their winding around a circular dimension. So, at first blush, it seems that open and closed strings behave completely differently under T-duality. With closer inspection, and by making use of the Dirichlet boundary conditions for open strings (the “D” in D-branes), Polchinski, Dai, Leigh, as well as Hořava, Green, and other researchers resolved this puzzle.

6. Proposals that have tried to circumvent the introduction of dark matter or dark energy have suggested that even the accepted behavior of gravity on large scales may differ from what Newton or Einstein would have thought, and in that way attempt to account for gravitational effects incompatible with solely the material we can see. As yet, these proposals are highly speculative and have little support, either experimental or theoretical.

7. The physicists who introduced this idea are S. Giddings and S. Thomas, and S. Dimopoulus and G. Landsberg.

8. Notice that the contraction phase of such a bouncing universe is not the same as the expansion phase run in reverse. Physical processes such as eggs splattering and candles melting would happen in the usual “forward” time direction during the expansion phases and would continue to do so during the subsequent contraction phase. That’s why entropy would increase during both phases.

9. The expert reader will note that the cyclic model can be phrased in the language of four-dimensional effective field theory on one of the three-branes, and in this form it shares many features with more familiar scalar-field-driven inflationary models. When I say “radically new mechanism,” I am referring to the conceptual description in terms of colliding branes, which in and of itself is a striking new way of thinking about cosmology.

10. Don’t get confused on dimension counting. The two three-branes, together with the space interval between them, have four dimensions. Time brings it to five. That leaves six more for the Calabi-Yau space.

11. An important exception, mentioned at the end of this chapter and discussed in further detail in Chapter 14, has to do with inhomogeneities